 Name _____ Date _____

Plague! How Disease Spreads

What can the Black Death teach us — and why did some people survive?

In this activity you'll run a disease spread simulation, collect data, and use it to explain how infectious diseases spread and why some people are more likely to survive.

PART A — Before You Simulate — Predictions

You are about to run the Disease Spread Simulation (New Visions). Open it from your unit page. Before you click anything, answer these prediction questions.

1. You start with 1 infected person in a town of 100. After 10 days, predict: how many people will be infected?

2. What do you think will happen to the spread if you make the town MORE crowded (higher population density)?

3. What do you think will happen if you give the population a vaccine?

PART B — Run the Simulation — Collect Data

Instructions: *Go to the Disease Spread Simulation on your unit page. Run THREE rounds. For each round, record the settings and your results in the table below. Try to change only ONE variable at a time (keep the disease the same).*

	Round 1	Round 2	Round 3
Disease chosen			
Population size			
Vaccination? (Y/N)			
Crowding level			
Days until 50% infected			
Total infected at end			
Total NOT infected			

PART C — Analyze Your Data

4. Which variable had the BIGGEST effect on how fast the disease spread? Use your data to support your answer.

5. Look at your Round 1 results. Compare how many people got infected vs. how many stayed healthy. Why didn't EVERYONE get infected?

6. What happened to the spread when you added vaccination? Explain using the idea of variation in immune systems.

PART D Connect to the Black Death

> ***Remember:*** *From 1347–1351, the Black Death killed an estimated 1 in 3 people in Europe. It was caused by the bacterium Yersinia pestis, carried by fleas on rats. It spread fastest in crowded cities with poor sanitation — dirty water, no hand washing, and rats everywhere.*

7. Using what you found in the simulation, explain why the plague spread so quickly through medieval European cities.

8. Not everyone who was exposed to the plague died. Scientists studying old skeletons found that some people had immune system gene variants that helped them survive. Connect this to the simulation: what does this remind you of?

9. The plague still exists today. The U.S. gets about 5 cases per year, mostly in the Southwest. But it's no longer a death sentence — why not? (Hint: think about what we have now that people in 1347 didn't.)

PART E Three Things That Make Disease Spread Faster

Fill in the table using your simulation results and class notes.

Factor	What it means	Black Death example	Modern example

Crowding			
Mixing / Travel			
Poor hygiene			

■ **Exit Ticket** In your own words, explain why the same disease might spread differently in a crowded city vs. a small rural village. Use at least 2 vocabulary words.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Simulation: simulations.newvisions.org/disease_transmission.html · Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html